

Amazon Aurora MySQL Serverless Cluster Setup

To Begin with the Lab

Summary of the Lab

In this video, an **Amazon Aurora Serverless** database cluster was created from scratch using the **Amazon RDS** console. The cluster was configured as **Aurora MySQL-compatible** and filtered to show versions that support both the **global database** feature. The setup followed the same core steps used for provisioned Aurora clusters, including setting the database identifier, master credentials, VPC, subnet group, security group, backups, and encryption. Instead of selecting a fixed instance class, the cluster used **Serverless v2** and Aurora Capacity Units to scale automatically based on demand. The lab confirmed that creating a serverless Aurora cluster is nearly the same as creating a provisioned cluster, with the main difference being automatic scaling.

Understanding Amazon Aurora Provisioned Clusters

An **Amazon Aurora provisioned cluster** is a database deployment where you choose and manage the compute capacity for each instance in the cluster. It exists so you can run a high-availability database with a writer instance for all writes and one or more reader instances for read scaling and failover support. Aurora clusters also provide separate writer and reader endpoints, which makes application routing easier because write traffic and read traffic can be sent to different targets. This is useful for production workloads that need strong availability, controlled performance, and the option to grow later into features such as global databases or additional replicas.

Example:

A web application stores orders in Aurora MySQL. The application sends INSERT and UPDATE operations to the writer endpoint, while reporting dashboards send SELECT queries to the reader endpoint. If the writer fails, Aurora can fail over to a replica in another Availability Zone and keep the database available.

Lab Steps

Step 1 – Start the Aurora Cluster Creation

- Sign in to the AWS Management Console.
- Open **Amazon RDS**.
- Click **Create database**.
- Select **Standard create**.
- Choose **Aurora MySQL compatible**.

Create database [Info](#)

Availability zone group [Info](#)
 Availability zone group
 Select the availability zone where you want to create your database.
 US East (N. Virginia)

Engine options

Engine type [Info](#)

☒ Aurora (MySQL Compatible)

☐ Aurora (PostgreSQL Compatible)

☐ MySQL

☐ PostgreSQL

☐ MariaDB

☐ Oracle

☐ Microsoft SQL Server

☐ IBM Db2

Choose a database creation method

☒ Full configuration

☐ Easy create

- Choose Database creation method as **Full configuration**.
- Choose **Production** as the template.
- Choose **Aurora Serverless Cluster**.
- And for Lab purpose, I have taken the lowest **ACU** for their minimum and maximum capacity.

Choose a database creation method

☒ Full configuration

☐ Easy create

Templates

Choose a sample template to meet your use case.

☒ Production

☐ Dev/Test

Cluster scalability type [Info](#)

☒ Aurora serverless

☐ Provisioned

Capacity range [Info](#)

Database capacity is measured in Aurora Capacity Units (ACUs). 1 ACU provides 2 GiB of memory and corresponding compute and networking. The platform version has been updated to Version 4 offering scaling up to 256 ACUs, and performance improvement up to 30% compared to version 3.

Minimum capacity (ACUs) (0 GiB)

Maximum capacity (ACUs) (2 GiB)

Pause after inactivity [Info](#)

Clusters with a minimum capacity setting of 0 ACUs can be paused during inactivity. Specify the amount of time the DB instance can be idle before pausing. For more information, see [Pause Aurora Serverless DB Instances](#).

Step 2 – Configure the Database Cluster

- Choose the compatible Engine Version based on your MySQL Version.
- Enter a database cluster identifier.
- Use the default master username.

Settings

Engine version
View the engine versions based on your selected database and cluster scalability type.
Aurora MySQL 3.12.0 (compatible with MySQL 8.0.44)

☐ Enable RDS Extended Support [Info](#)
Amazon RDS Extended Support is a [paid offering](#). By selecting this option, you consent to being charged for this offering if you are running your database major version past the RDS end of standard support date for that version. Check the end of standard support date for your major version in the [Amazon Aurora documentation](#).

DB cluster identifier [Info](#)
Type a name for your DB cluster. The name must be unique across all DB clusters owned by your AWS account in the current AWS Region.
database-2
The DB cluster identifier is case-insensitive, but is stored as all lowercase (as in "mydbcluster"). Constraints: 1 to 63 alphanumeric characters or hyphens. First character must be a letter. Can't contain two consecutive hyphens. Can't end with a hyphen.

Credentials Settings

Master username [Info](#)
Type a login ID for the master user of your DB instance.
admin
1 to 32 alphanumeric characters. The first character must be a letter.

Credentials management [Info](#)
You can use AWS Secrets Manager or manage your master user credentials.
☐ Managed in AWS Secrets Manager
RDS generates a password for you and manages it throughout its lifecycle using AWS Secrets Manager.
☒ Self managed
Create your own password or have RDS create a password that you manage.

☐ Auto generate password
Amazon RDS can generate a password for you, or you can specify your own password.

Master password [Info](#)

- I will choose **Self Managed** option.
- Set the master password and choose the **Password authentication** method.

☐ Managed in AWS Secrets Manager
RDS generates a password for you and manages it throughout its lifecycle using AWS Secrets Manager.

☒ Self managed
Create your own password or have RDS create a password that you manage.

☐ Auto generate password
Amazon RDS can generate a password for you, or you can specify your own password.

Master password [Info](#)

Minimum constraints: At least 8 printable ASCII characters. Can't contain any of the following symbols: / ' * @
Password strength: **Weak**

Confirm master password [Info](#)

Additional credentials settings

Database authentication
Manage your database user's credentials. Password authentication is always active for your database engine. You can turn on additional authentication methods for your database below.

Database authentication options [Info](#)
☒ Password authentication
Authenticates using database passwords.
☐ Password and IAM database authentication
Authenticates using the database password and user credentials through AWS IAM users and roles.

Cluster storage configuration [Info](#)
Choose the storage configuration for the Aurora DB cluster that best fits your application's price predictability and price performance needs.

Configuration options
Database instance, storage, and I/O charges vary depending on the configuration. [Learn more](#)

- Choose **Aurora standard** storage.
- Create an Aurora replica in a different Availability Zone for failover support.

Cluster storage configuration [info](#)

Choose the storage configuration for the Aurora DB cluster that best fits your application's price predictability and price performance needs.

Configuration options

Database instance, storage, and I/O charges vary depending on the configuration. [Learn more](#)

☐ Aurora I/O-Optimized

- Predictable pricing for all applications. Improved price performance for I/O-intensive applications (I/O costs >25% of total database costs).
- No additional charges for read/write I/O operations. DB instance and storage prices include I/O usage.

☒ **Aurora Standard**

- Cost-effective pricing for many applications with moderate I/O usage (I/O costs <25% of total database costs).
- Pay-per-request I/O charges apply. DB instance and storage prices don't include I/O usage.

Availability & durability

Multi-AZ deployment [info](#)

☒ **Create an Aurora Replica or Reader node in a different AZ (recommended for scaled availability)**

☐ Don't create an Aurora Replica

Connectivity [info](#)

Compute resource

Choose whether to set up a connection to a compute resource for this database. Setting up a connection will automatically change connectivity settings so that the compute resource can connect to this database.

☒ **Don't connect to an EC2 compute resource**

Don't set up a connection to a compute resource for this database. You can manually set up a connection to a compute resource later.

☐ Connect to an EC2 compute resource

Set up a connection to an EC2 compute resource for this database.

Network type [info](#)

To use dual-stack mode, make sure that you associate an IPv6 CIDR block with a subnet in the VPC you specify.

Step 3 – Configure Connectivity and Security

- Select the VPC where the cluster will be created.
- Do not connect the cluster to an EC2 compute resource automatically.
- Select the VPC if you have created it but I will go with Default values here.
- Set **Public access** to **No** for lab purpose.

Connectivity [info](#)

Compute resource

Choose whether to set up a connection to a compute resource for this database. Setting up a connection will automatically change connectivity settings so that the compute resource can connect to this database.

☒ **Don't connect to an EC2 compute resource**

Don't set up a connection to a compute resource for this database. You can manually set up a connection to a compute resource later.

☐ Connect to an EC2 compute resource

Set up a connection to an EC2 compute resource for this database.

Network type [info](#)

To use dual-stack mode, make sure that you associate an IPv6 CIDR block with a subnet in the VPC you specify.

☒ **IPv4**

Your resources can communicate only over the IPv4 addressing protocol.

☐ Dual-stack mode

Your resources can communicate over IPv4, IPv6, or both.

Virtual private cloud (VPC) [info](#)

Choose the VPC. The VPC defines the virtual networking environment for this DB cluster.

Default VPC (vpc-099792461c7d0a616)

After a database is created, you can't change its VPC. Only VPCs with a corresponding DB subnet group are listed.

DB subnet group [info](#)

Choose the DB subnet group. The DB subnet group defines which subnets and IP ranges the DB cluster can use in the VPC that you selected.

default-vpc-099792461c7d0a616

Public access [info](#)

☐ Yes

RDS assigns a public IP address to the cluster. Amazon EC2 instances and other resources outside of the VPC can connect to your cluster. Resources inside the VPC can also connect to the cluster. Choose one or more VPC security groups that specify which resources can connect to the cluster.

☒ **No**

RDS doesn't assign a public IP address to the cluster. Only Amazon EC2 instances and other resources inside the VPC can connect to your cluster. Choose one or more VPC security groups that specify which resources can connect to the cluster.

- Select the existing security group if created otherwise go with **Default**.
- Leave **RDS Proxy** disabled for this lab.

aws [Search] [Alt+S] United States (N. Virginia) cloudriteshk (463646775279) cf-shashank1

Aurora and RDS > Databases > Create database

☒ No
RDS doesn't assign a public IP address to the cluster. Only Amazon EC2 instances and other resources inside the VPC can connect to your cluster. Choose one or more VPC security groups that specify which resources can connect to the cluster.

VPC security group (firewall) [Info](#)
Choose one or more VPC security groups to allow access to your database. Make sure that the security group rules allow the appropriate incoming traffic.

☒ Choose existing [Choose existing VPC security groups](#) ☐ Create new [Create new VPC security group](#)

Existing VPC security groups
[Choose one or more options](#)

default X

RDS Proxy
RDS Proxy is a managed, highly available database proxy that improves application scalability, resiliency, and security.

☐ Create an RDS Proxy [Info](#)
RDS Proxy requires an IAM role and a Secrets Manager secret for the proxy. RDS Proxy has additional costs. For more information, see [Amazon RDS Proxy pricing](#).

Certificate authority - optional [Info](#)
Using a server certificate provides an extra layer of security by validating that the connection is being made to an Amazon database. It does so by checking the server certificate that is automatically installed on all databases that you provision.

rdc-ca-rsa2048-g1 (default)
Expiry: May 26, 2061

If you don't select a certificate authority, RDS chooses one for you.

► Additional configuration

Tags - optional
A tag consists of a case-sensitive key-value pair.

- **Keep Performance Insights enabled.**

aws [Search] [Alt+S] United States (N. Virginia) cloudriteshk (463646775279) cf-shashank1

Aurora and RDS > Databases > Create database

Monitoring [Info](#)
Choose monitoring tools for this database. Database Insights provides a combined view of Performance Insights and Enhanced Monitoring for your fleet of databases.

☐ Database Insights - Advanced
• Retains 15 months of performance history
• Fleet-level monitoring
• Integration with CloudWatch Application Signals

☒ Database Insights - Standard

CloudWatch Database Insights pricing is separate from RDS monthly estimates. See [Amazon CloudWatch pricing](#).

Performance Insights
☒ Enable Performance insights
With Performance Insights dashboard, you can visualize the database load on your Amazon RDS DB instance load and filter the load by waits, SQL statements, hosts, or users.

Retention period
7 days

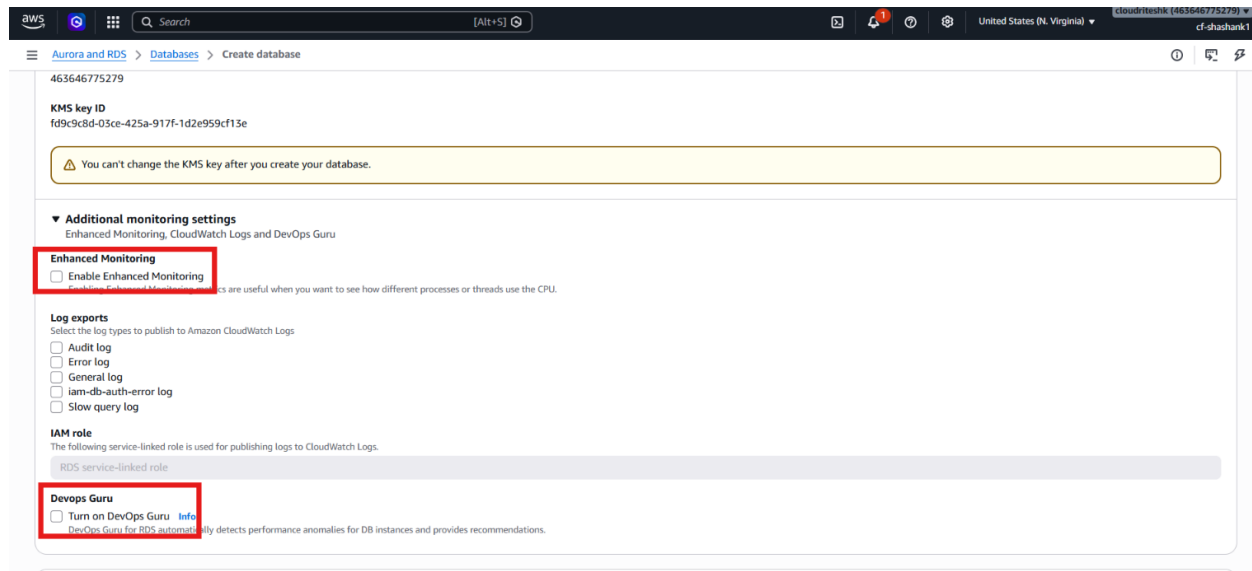
AWS KMS key [Info](#)
arn:aws:kms:us-east-1:463646775279:key/fd9c9c8d-03ce-425a-917f-1d2e959cf13e
Example: arn:aws:kms:<region>:<accountID>:key/<key-id>

Alias
(default) aws/rds

Account
463646775279

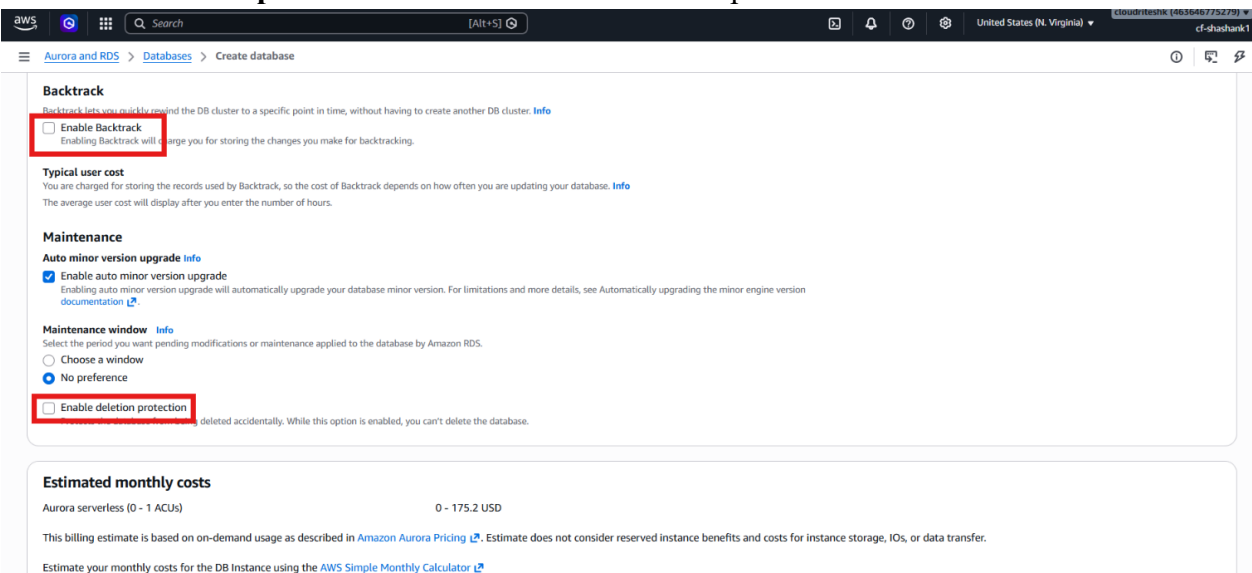
KMS key ID
fd9c9c8d-03ce-425a-917f-1d2e959cf13e

- **Disable Enhanced Monitoring and Devops Guru only for lab purpose otherwise they will be automatically enabled.**



Step 4 – Configure Advanced Options

- Keep **Backtrack** disabled for this lab.
- Set the maintenance window if needed.
- Leave **Deletion protection** disabled for easier cleanup.



- Click **Create database**.

Step 5 – Wait for Cluster Creation

- Wait while Aurora creates the cluster.
- Confirm that the cluster status becomes available.
- Verify that the cluster contains one writer instance and one reader instance.
- Confirm that the writer handles write operations.
- Confirm that the reader is available for read operations.

Step 6 – Verify Endpoints and Cluster Details

- Open the cluster details page.
- Scroll to **Connectivity and security**.
- Check the **Configuration** section.

The screenshot shows the AWS Aurora console with the 'database-2' cluster selected. The 'database-2-instance-1' instance is highlighted in the instance list. The 'Configuration' tab is active, showing details for the instance. The 'Instance configuration' section is expanded, and the 'Instance type' is set to 'Aurora serverless', which is highlighted with a red box. Other configuration details include DB instance ID, engine version, RDS Extended Support, DB name, and option groups.

DB identifier	Status	Role	Engine	Region	Size	Recommendations	CPU
database-2	Available	Regional cluster	Aurora MySQL	us-east-1	2 instances	2 Informational	-
database-2-instance-1	Available	Writer instance	Aurora MySQL	us-east-1d	Aurora serverless	-	-
database-2-instance-1-reader	Creating	Reader instance	Aurora MySQL	us-east-1a	Aurora serverless	-	-

Instance Configuration:

- DB instance ID: database-2-instance-1
- Engine version: 8.0.mysql_aurora.3.12.0
- RDS Extended Support: Disabled
- DB name: -
- Option groups: -
- Instance type: **Aurora serverless**
- Minimum capacity: 0 ACUs (0 GiB)
- Maximum capacity: 1 ACUs (2 GiB)
- Platform version: 4 (Version 4 offering scaling up to 256 ACUs, and performance improvement up to 30% compared to version 3)

Primary storage:

- Encryption key: AWS managed KMS key
- Storage type: -

Monitoring:

- Monitoring type: Database Insights - Standard
- Performance Insights: Disabled
- Enhanced Monitoring: Disabled

- Confirm that the cluster is serverless.

The screenshot shows the AWS Aurora console with the 'database-2' cluster selected. The 'Configuration' tab is active, showing details for the cluster. The 'Availability' section is expanded, and the 'Multi-AZ' option is set to '2 Zones', which is highlighted with a red box. Other configuration details include DB cluster role, engine version, RDS Extended Support, resource ID, cluster storage configuration, capacity type, authentication, encryption, and monitoring.

DB identifier	Status	Role	Engine	Region	Size	Recommendations	CPU
database-2	Available	Regional cluster	Aurora MySQL	us-east-1	2 instances	2 Informational	-
database-2-instance-1	Available	Writer instance	Aurora MySQL	us-east-1d	Aurora serverless	-	-
database-2-instance-1-reader	Creating	Reader instance	Aurora MySQL	us-east-1a	Aurora serverless	-	-

Database Configuration:

- DB cluster role: Regional cluster
- Engine version: 8.0.mysql_aurora.3.12.0
- RDS Extended Support: Disabled
- Resource ID: cluster-HHQC6534QHFLQXXMDGW6CNNE-OA
- Cluster storage configuration: Aurora Standard
- Capacity type: Provisioned
- Local read replica write forwarding: Disabled
- DB cluster ID: database-2
- DB cluster parameter group: default:aurora-mysql8.0
- Deletion protection: Disabled
- Limitless Database: Disabled

Authentication:

- IAM DB authentication: Not enabled
- Kerberos authentication: Not enabled
- Master username: admin
- Master password: *****

Encryption:

- Encryption key: AWS managed KMS key

Monitoring:

- Monitoring type: Database Insights - Standard
- Performance Insights: Disabled
- Enhanced Monitoring: Disabled

Availability:

- Multi-AZ: **2 Zones**

- Confirm that the cluster spans two Availability Zones.

Verification

- The **Amazon Aurora MySQL** cluster was created successfully.
- The cluster used the **serverless** deployment model.
- A writer instance and a reader instance were created.
- The writer endpoint and reader endpoint were available.

- The cluster was spread across two Availability Zones.
- Advanced options such as backups, encryption, and failover priority were reviewed successfully.